Aug 14 2025 BDBP106: Linux and Python programming Lab 8

Learning goals: Basics of VIM editor
NOTE: Save screenshots of each exercise, and upload your work to your github account as Lab8.pdf and the modified data file by end of Thursday Aug 14.
EXERCISES Start a fresh terminal for this Lab and make a new directory called ’Lab8’. Copy the ‘Braincancer.csv’ file to this directory. At the end of the lab, rename the data file as ‘Braincancer Lab8.csv’ and upload this to github as well.

(1) Open the data file using the VIM editor.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~$ cd Downloads
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ cd lab8
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads/lab8$ vim BrainCancer
```

(2) Move to the beginning of the file and insert the text: ”VIM challenge begins!” on a new line at the very top.

```
"VIM Challenge begins!"
,sex,diagnosis,loc,ki,gtv,stereo,status,time
1,Female,Meningioma,Infratentorial,90,6.11,SRS,0,57.64
2,Male,HG glioma,Supratentorial,90,19.35,SRT,1,8.98
```

(3) Find the word ‘Meningioma’ and replace it with ‘LG glioma’

```
19,Male,Meningioma,Supratentorial,80,13.49,SRS,1,6.92
20,Female,Meningioma,Supratentorial,90,2.5,SRT,0,30.16
21,Female,Meningioma,Supratentorial,80,2.82,SRS,0,24.39
:%s/Meningioma/LG glioma/gc
```

```
86,Male,LG glioma,Supratentorial,90,2.5,SRT,0,32.82
87,Male,LG glioma,Supratentorial,90,2.02,SRS,0,42.07
88,Male,Other,Infratentorial,80,0.11,SRT,0,13.9
42 substitutions on 42 lines                                    89
```

(4) Delete the entire line containing the time column entry ‘34.66’

```
8,Male,LG glioma,Supratentorial,80,12.37,SRT,0,42.1
9,Female,LG glioma,Supratentorial,70,12.16,SRT,0,34.66
10,Male,HG glioma,Supratentorial,100,2.53,SRT,0,11.48
11,Male,LG glioma,Supratentorial,80,0.14,SRT,1,35.93
12,Female,LG glioma,Infratentorial,90,6.54,SRS,0,34.26
13,Female,LG glioma,Infratentorial,90,0.63,SRS,0,32.98
14,Male,NA,Supratentorial,90,6.38,SRT,0,50.85
15,Female,LG glioma,Supratentorial,60,9.18,SRT,0,41.44
16,Female,HG glioma,Supratentorial,70,11.38,SRS,1,7.05
17,Female,Other,Infratentorial,60,24,SRT,1,6.82
18,Male,HG glioma,Supratentorial,90,10.8,SRT,0,82.56
19,Male,LG glioma,Supratentorial,80,13.49,SRS,1,6.92
20,Female,LG glioma,Supratentorial,90,2.5,SRT,0,30.16
21,Female,LG glioma,Supratentorial,80,2.82,SRS,0,24.39
22,Male,HG glioma,Supratentorial,70,14.44,SRT,1,14
23,Female,Other,Infratentorial,80,2.11,SRS,0,10.49
24,Female,LG glioma,Infratentorial,100,2.13,SRS,1,51.02
25,Female,LG glioma,Supratentorial,70,6.48,SRT,1,33.41
26,Male,LG glioma,Supratentorial,90,4.23,SRT,1,25.02
27,Male,Other,Supratentorial,60,34.64,SRT,1,11.57
28,Male,HG glioma,Supratentorial,70,33.69,SRT,1,0.07
29,Male,LG glioma,Supratentorial,60,3.81,SRT,0,36.1
30,Female,LG glioma,Supratentorial,90,4.72,SRS,0,65.02
31,Female,LG glioma,Supratentorial,80,0.85,SRS,1,6.1
32,Male,LG glioma,Supratentorial,90,2.56,SRS,0,44.39
33,Female,Other,Infratentorial,70,13.45,SRT,1,10.82
34,Male,Other,Infratentorial,80,6.81,SRS,0,57.11
35,Female,LG glioma,Supratentorial,90,7.3,SRT,0,5.51
36,Female,Other,Supratentorial,70,14.26,SRT,0,7.18
37,Female,LG glioma,Supratentorial,80,6.6,SRT,0,14.75
38,Male,HG glioma,Supratentorial,90,9.95,SRT,1,6.23
39,Male,Other,Infratentorial,80,12.51,SRT,1,29.7
40,Female,LG glioma,
41,Female,LG glioma,
42,Male,HG glioma,Su
43,Female,LG glioma,
44,Male,LG glioma,Su
45,Male,Other,Infrat
46,Male,LG glioma,Su
47,Male,HG glioma,Su
48,Female,LG glioma,
49,Male,LG glioma,Su
50,Female,LG glioma,
51,Female,HG glioma,
52,Female,LG glioma,
/34.66
```

```
8,Male,LG glioma,Supratentorial,80,12.37,SRT,0,42.1
10,Male,HG glioma,Supratentorial,100,2.53,SRT,0,11.48
11,Male,LG glioma,Supratentorial,80,0.14,SRT,1,35.93
12,Female,LG glioma,Infratentorial,90,6.54,SRS,0,34.26
13,Female,LG glioma,Infratentorial,90,0.63,SRS,0,32.98
```

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(5) Copy the current line and paste it two lines below. Show the screenshot before and after pasting with file line numbers to prove that you did it correctly.

```
 9 7,Male,LG glioma,Supratentorial,80,3.19,SRT,0,55.8
10 8,Male,LG glioma,Supratentorial,80,12.37,SRT,0,42.1
11 10,Male,HG glioma,Supratentorial,100,2.53,SRT,0,11.48
12 11,Male,LG glioma,Supratentorial,80,0.14,SRT,1,35.93
13 12,Female,LG glioma,Infratentorial,90,6.54,SRS,0,34.26
14 13,Female,LG glioma,Infratentorial,90,0.63,SRS,0,32.98
```

```
10 8,Male,LG glioma,Supratentorial,80,12.37,SRT,0,42.1
11 10,Male,HG glioma,Supratentorial,100,2.53,SRT,0,11.48
12 11,Male,LG glioma,Supratentorial,80,0.14,SRT,1,35.93
13 9,Female,LG glioma,Supratentorial,70,12.16,SRT,0,34.66
14 12,Female,LG glioma,Infratentorial,90,6.54,SRS,0,34.26
```

(6) Move to the 11th line of this file and insert ”Hello from line 11!”

```
 9 7,Male,LG glioma,Supratentorial,80,3.19,SRT,0,55.8
10 8,Male,LG glioma,Supratentorial,80,12.37,SRT,0,42.1
11 "Hello from line 11!"
12 11,Male,LG glioma,Supratentorial,80,0.14,SRT,1,35.93
13 9,Female,LG glioma,Supratentorial,70,12.16,SRT,0,34.66
14 12,Female,LG glioma,Infratentorial,90,6.54,SRS,0,34.26
```

(7) Search for the word ‘SRT’ and replace it with ‘SRS’ (all occurences in the file)

```
49,Male,LG glioma,Supratentorial,70,0.97,SRT,1,1.41
50,Female,LG glioma,Supratentorial,80,0.19,SRS,0,11.51
51,Female,HG glioma,Supratentorial,90,0.04,SRT,0,31.67
52,Female,LG glioma,Infratentorial,90,9.24,SRT,0,26.85
:%s/SRT/SRS/gc
```

```
85,Male,HG glioma,Supratentorial,80,19.81,SRS,1,6.39
86,Male,LG glioma,Supratentorial,90,2.5,SRS,0,32.82
87,Male,LG glioma,Supratentorial,90,2.02,SRS,0,42.07
88,Male,Other,Infratentorial,80,0.11,SRS,0,13.9
64 substitutions on 64 lines
```

(8) Yank the first 3 lines of the file and paste them at the bottom

```
 90 88,Male,Other,Infratentorial,80,0.11,SRS,0,13.9
 91 "VIM Challenge begins!"
 92 ,sex,diagnosis,loc,ki,gtv,stereo,status,time
 93 1,Female,LG glioma,Infratentorial,90,6.11,SRS,0,57.64
3 more lines
```

(9) Visually select this entire instructions list and indent it by two spaces.
Crtl v takes to visual block
shift g -----yank---- paste--- replace with space

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```
1  "VIM Challenge begins!"
2  ,sex,diagnosis,loc,ki,gtv,stereo,status,time
3  1,Female,LG glioma,Infratentorial,90,6.11,SRS,0,57.64
4  2,Male,HG glioma,Supratentorial,90,19.35,SRS,1,8.98
5  3,Female,LG glioma,Infratentorial,70,7.95,SRS,0,26.46
6  4,Female,LG glioma,Supratentorial,80,7.61,SRS,1,47.8
7  5,Male,HG glioma,Supratentorial,90,5.06,SRS,1,6.3
8  6,Female,LG glioma,Supratentorial,80,4.82,SRS,0,52.75
9  7,Male,LG glioma,Supratentorial,80,3.19,SRS,0,55.8
10 8,Male,LG glioma,Supratentorial,80,12.37,SRS,0,42.1
11 "Hello from line 11!"
12 11,Male,LG glioma,Supratentorial,80,0.14,SRS,1,35.93
13 9,Female,LG glioma,Supratentorial,70,12.16,SRS,0,34.66
14 12,Female,LG glioma,Infratentorial,90,6.54,SRS,0,34.26
15 13,Female,LG glioma,Infratentorial,90,0.63,SRS,0,32.98
16 14,Male,NA,Supratentorial,90,6.38,SRS,0,50.85
17 15,Female,LG glioma,Supratentorial,60,9.18,SRS,0,41.44
18 16,Female,HG glioma,Supratentorial,70,11.38,SRS,1,7.05
19 17,Female,Other,Infratentorial,60,24,SRS,1,6.82
20 18,Male,HG glioma,Supratentorial,90,10.8,SRS,0,82.56
21 19,Male,LG glioma,Supratentorial,80,13.49,SRS,1,6.92
22 20,Female,LG glioma,Supratentorial,90,2.5,SRS,0,30.16
23 21,Female,LG glioma,Supratentorial,80,2.82,SRS,0,24.39
24 22,Male,HG glioma,Supratentorial,70,14.44,SRS,1,14
25 23,Female,Other,Infratentorial,80,2.11,SRS,0,10.49
```

(10) Go to the 50th line number, and insert the following text: “ The cat sat on the mat. Another cat was watching. The cat did not move.” Replace all occurences of the word “cat” with ”tiger”.

```
47  45,Male,Other,Infratentorial,90,3.12,SRS,0,18.95
48  46,Male,LG glioma,Supratentorial,60,7.09,SRS,1,31.25
49  47,Male,HG glioma,Supratentorial,80,29.27,SRS,0,5.15
50  The cat sat on the mat. Another cat was watching. The cat did not move.
51  49,Male,LG glioma,Supratentorial,70,0.97,SRS,1,1.41
```

```
60  58,Male,HG glioma,Supratentorial,80,3.75,SRS,1,19.9
61  59,Female,Other,Supratentorial,80,11.83,SRS,1,22.03
:%s/cat/tiger/gc
```

```
49  47,Male,HG glioma,Supratentorial,80,29.27,SRS,0,5.15
50  The tiger sat on the mat. Another tiger was watching. The tiger did not move.
51  49,Male,LG glioma,Supratentorial,70,0.97,SRS,1,1.41
```

(11) Insert the text “This is an important reminder for you.” at 70th line number. Then change the word ‘important’ to ‘critical’. Describe the sequence of keystrokes used in full.

Navigate to 70th line
press ---i insert the text
:%s/important/critical

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```
69   67,Female,HG glioma,Supratentorial,80,7.59,SRS,1,4.56
70   This an important remainder for you
71   69,Female,LG glioma,Supratentorial,80,6.93,SRS,0,10.1
72   70,Female,LG glioma,Supratentorial,70,3.63,SRS,0,32.82
73   71,Male,LG glioma,Supratentorial,70,8.45,SRS,0,19.41
74   72,Male,LG glioma,Supratentorial,80,20.93,SRS,1,31.15
75   73,Male,LG glioma,Supratentorial,90,2.64,SRS,0,20.13
76   74,Female,HG glioma,Supratentorial,80,0.19,SRS,1,11.02
77   75,Male,Other,Supratentorial,100,24.91,SRS,0,19.74
:%s/important/critical
```

```
69   67,Female,HG glioma,Supratentorial,80,7.59,SRS,1,4.56
70   This an critical remainder for you
```

(12) Note down the current line number, jump to the end of the file and jump back to this line without scrolling through every line. Describe the sequence of keystrokes used in full.
Shift g jumps to end of line
<line number> 90 shift g

```
90   88,Male,Other,Infratentorial,80,0.11,SRS,0,13.9
91   "VIM Challenge begins!"
92   ,sex,diagnosis,loc,ki,gtv,stereo,status,time
93   1,Female,LG glioma,Infratentorial,90,6.11,SRS,0,57.64
~
                                                    90,3
```

```
89   87,Male,LG glioma,Supratentorial,90,2.02,SRS,0,42.07
90   88,Male,Other,Infratentorial,80,0.11,SRS,0,13.9
91   "VIM Challenge begins!"
92   ,sex,diagnosis,loc,ki,gtv,stereo,status,time
93   1,Female,LG glioma,Infratentorial,90,6.11,SRS,0,57.64
~
                                                    93,3
```

```
89   87,Male,LG glioma,Supratentorial,90,2.02,SRS,0,42.07
90   88,Male,Other,Infratentorial,80,0.11,SRS,0,13.9
91   "VIM Challenge begins!"
92   ,sex,diagnosis,loc,ki,gtv,stereo,status,time
93   1,Female,LG glioma,Infratentorial,90,6.11,SRS,0,57.64
~
                                                    90,3
```

(13) Add the following sentence at the end of the file: ”The quick brown fox (who was very quick) jumps over the lazy dog.” Delete the text inside the parentheses (including the parentheses) in the following text. Describe the sequence of keystrokes used in full.
Visual mode at the parentheses select the text and parentheses and x for delete

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```
87,Male,LG glioma,Supratentorial,90,2.02,SRS,0,42.07
88,Male,Other,Infratentorial,80,0.11,SRS,0,13.9
"VIM Challenge begins!"
,sex,diagnosis,loc,ki,gtv,stereo,status,time
The quick brown fox (who was very quick) jumps over the lazy dog
~
-- INSERT --                                                    93,67
```

```
85,Male,HG glioma,Supratentorial,80,19.81,SRS,1,6.39
86,Male,LG glioma,Supratentorial,90,2.5,SRS,0,32.82
87,Male,LG glioma,Supratentorial,90,2.02,SRS,0,42.07
88,Male,Other,Infratentorial,80,0.11,SRS,0,13.9
"VIM Challenge begins!"
,sex,diagnosis,loc,ki,gtv,stereo,status,time
The quick brown fox jumps over the lazy dog
```

(14) Go to the last line of the file and append: ”– end of challenge”

```
86,Male,LG glioma,Supratentorial,90,2.5,SRS,0,32.82
87,Male,LG glioma,Supratentorial,90,2.02,SRS,0,42.07
88,Male,Other,Infratentorial,80,0.11,SRS,0,13.9
"VIM Challenge begins!"
,sex,diagnosis,loc,ki,gtv,stereo,status,time
The quick brown fox jumps over the lazy dog"-end of challenge"
```

(15) In the remaining exercises, we are going to use Vim’s search, replace, visual modes and text objects. Open the file vim_cleanup_exercises.txt and do the following. Describe the sequence of keystrokes fully for each problem.

(i) Fix spacing issues in the text in the lines 1-3.

```
1 This line has too many spaces.
2 Replace tabs with single spaces.
3 This line starts with too many spaces.
```

:%s/ / /g
:%s/ / /g
3rd line x
(ii) Fix any capitalization and typographical errors in lines 5-8.

```
5 The quick brown fox jumps over the lazy dog.
6 VIM is a amazing text editor.
7 I enjoy learning vim.
8 Python is fun but also challening.
```

Substitue with correct spellings
(iii) Correct punctuation errors in lines 10-13.
insert punctuation with i

```
10 What is your name?
11 Yes,I like apples.
12 Wait this is too many dots.
13 ----------------------------------------------
```

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(iv) Remove unwanted words in lines 15-17.

```
---------------------------------------------
I really just want to say that this is fine.
Please delete the word anywhere it appears.
I will fix this line.
---------------------------------------------
Name      Age City
```

:s/literally//
:s/totally//
:%s/TODO//g

(v) Fix alignment issues in lines 19-22.
insert i and navigate with j k l h keys and x for delete

```
---------------------------------------------
Name      Age City
Alice     24  New York
Bob       32  Los Angeles
Charlie   19  Chicago
---------------------------------------------
```